CHSH Bell Inequality Violation

Juan David de los Rios Mahecha
Departamento de fisica, universidad distrital Francisco Jose de Caldas
(Dated: April 23, 2026)

I. THEORETICAL FRAMEWORK

A. The CHSH Game: Classical Formulation

We consider a cooperative game between two spatially separated players, Alice and Bob, supervised by a referee. Once the game begins, the players are forbidden from communicating with one another. The referee independently and uniformly at random selects binary inputs $x \in \{0,1\}$ for Alice and $y \in \{0,1\}$ for Bob. Upon receiving their respective inputs, the players must produce binary outputs $a \in \{0,1\}$ and $b \in \{0,1\}$ satisfying the winning condition:

$$a \oplus b = x \cdot y, \tag{1}$$

where $\oplus$ denotes addition modulo 2 and $\cdot$ denotes ordinary multiplication. Table I summarizes the four possible input configurations and the corresponding required relationship between outputs.

TABLE I: Winning conditions of the CHSH game for each input pair (x, y).

x	y	$x \cdot y$	Required condition on a, b
0	0	0	$a = b$
0	1	0	$a = b$
1	0	0	$a = b$
1	1	1	$a \neq b$

It follows immediately from Table I that the winning condition requires equal outputs in three out of four cases, and strictly different outputs in the single case $(x, y) = (1, 1)$. This asymmetry is at the heart of the classical limitation established in Section I E.

B. Algebraic Reformulation: From Boolean to Sign Variables

The winning condition (1) is naturally expressed in Boolean algebra over $\{0,1\}$. However, connecting this formulation to the standard framework of Bell inequalities requires a change of representation to the sign algebra over $\{-1,+1\}$, in which correlations acquire a particularly transparent analytical form amenable to rigorous bounding arguments.

1. Change of Variables

We introduce signed variables via the invertible linear map:

$$\tilde{a} = (-1)^a, \qquad \tilde{b} = (-1)^b, \tag{2}$$

so that $a = 0 \mapsto \tilde{a} = +1$ and $a = 1 \mapsto \tilde{a} = -1$, and analogously for $\tilde{b}$. Under this map, the Boolean equality $a = b$ is equivalent to $\tilde{a}\tilde{b} = +1$, while the Boolean inequality $a \neq b$ is equivalent to $\tilde{a}\tilde{b} = -1$.

2. Algebraic Translation of the XOR Operation

The XOR operation satisfies the following fundamental identity under the map (2):

$$(-1)^{a \oplus b} = (-1)^a \cdot (-1)^b = \tilde{a}\,\tilde{b}, \tag{3}$$

which holds because addition modulo 2 in the exponent corresponds to multiplication of signs. Applying (3) to the winning condition (1), we obtain the equivalent signed formulation:

$$\tilde{a}\,\tilde{b} = (-1)^{x \cdot y}. \tag{4}$$

Evaluating the right-hand side explicitly for each input pair yields:

$$(x, y) = (0, 0): \quad (-1)^{0 \cdot 0} = (-1)^0 = +1, \tag{5}$$
$$(x, y) = (0, 1): \quad (-1)^{0 \cdot 1} = (-1)^0 = +1, \tag{6}$$
$$(x, y) = (1, 0): \quad (-1)^{1 \cdot 0} = (-1)^0 = +1, \tag{7}$$
$$(x, y) = (1, 1): \quad (-1)^{1 \cdot 1} = (-1)^1 = -1. \tag{8}$$

The winning condition thus requires $\tilde{a}\tilde{b} = +1$ in three of the four cases, and $\tilde{a}\tilde{b} = -1$ exclusively when $x = y = 1$.

C. The Correlation Function $E(x, y)$

1. Definition

For a given strategy, we define the correlation function $E(x, y)$ as the expectation value of the signed product $\tilde{a}\tilde{b}$ conditioned on the input pair (x, y):

$$E(x,y) \;=\; \mathbb{E}\big[\tilde{a}\,\tilde{b} \;\big|\; x,y\big] \;=\; \sum_{a,b\,\in\,\{0,1\}} P(a,b \mid x,y)\,(-1)^{a+b}. \tag{9}$$

For a deterministic strategy, $P(a,b \mid x,y) \in \{0,1\}$, and therefore $E(x,y) \in \{-1,+1\}$. For probabilistic mixed strategies, $E(x,y) \in [-1,+1]$ by convexity. The correlation function provides a continuous measure of the degree to which the outputs of Alice and Bob are aligned: $E(x,y) = +1$ indicates perfect correlation, $E(x,y) = -1$ indicates perfect anticorrelation, and $E(x,y) = 0$ indicates statistical independence.

2. *Explicit Computation for the Optimal Classical Strategy*

Without loss of generality, consider the deterministic strategy $a(x) = 0$ for all $x \in \{0,1\}$ and $b(y) = 0$ for all $y \in \{0,1\}$. Under this strategy, $\tilde{a} = +1$ and $\tilde{b} = +1$ for all inputs, so that $\tilde{a}\tilde{b} = +1$ universally. The correlation functions are therefore:

$$E(0,0) = +1, \tag{10}$$
$$E(0,1) = +1, \tag{11}$$
$$E(1,0) = +1, \tag{12}$$
$$E(1,1) = +1. \tag{13}$$

This strategy satisfies the winning condition for the inputs $(0,0)$, $(0,1)$, and $(1,0)$, where equal outputs are required, but fails for $(1,1)$, where different outputs are required. It constitutes the best achievable classical performance, as demonstrated rigorously in Section I E.

D. The CHSH Parameter

The CHSH parameter is defined as the following signed linear combination of correlation functions:

$$S \;=\; E(0,0) + E(0,1) + E(1,0) - E(1,1). \tag{14}$$

The sign structure in (14) is not arbitrary: it reflects precisely the asymmetry of the winning condition (1), assigning a positive weight to the three cases requiring correlation and a negative weight to the single case requiring anticorrelation. For the optimal classical strategy computed above:

$$S_{\text{classical}} = (+1) + (+1) + (+1) - (+1) = +2. \tag{15}$$

E. The Classical Bound: $|S| \leq 2$

[Classical CHSH Bound] For any local deterministic strategy, the CHSH parameter satisfies $|S| \leq 2$. By convexity, this bound extends to all classical probabilistic strategies.

Let $\tilde{a}(x) \in \{-1,+1\}$ and $\tilde{b}(y) \in \{-1,+1\}$ denote the signed outputs of Alice and Bob under a deterministic local strategy. Since the strategy is deterministic, the correlation functions factorize as:

$$E(x,y) = \tilde{a}(x)\,\tilde{b}(y). \tag{16}$$

Substituting into (14):

$$S = \tilde{a}(0)\tilde{b}(0) + \tilde{a}(0)\tilde{b}(1) + \tilde{a}(1)\tilde{b}(0) - \tilde{a}(1)\tilde{b}(1). \tag{17}$$

Factoring by Alice's outputs:

$$S = \tilde{a}(0)\big[\tilde{b}(0) + \tilde{b}(1)\big] + \tilde{a}(1)\big[\tilde{b}(0) - \tilde{b}(1)\big]. \tag{18}$$

Since $\tilde{b}(0), \tilde{b}(1) \in \{-1,+1\}$, there are exactly two cases:

- **Case 1:** $\tilde{b}(0) = \tilde{b}(1)$. Then $\tilde{b}(0) + \tilde{b}(1) = \pm 2$ and $\tilde{b}(0) - \tilde{b}(1) = 0$, yielding $S = \pm 2\,\tilde{a}(0)$, so $|S| = 2$.
- **Case 2:** $\tilde{b}(0) \neq \tilde{b}(1)$. Then $\tilde{b}(0) + \tilde{b}(1) = 0$ and $\tilde{b}(0) - \tilde{b}(1) = \pm 2$, yielding $S = \pm 2\,\tilde{a}(1)$, so $|S| = 2$.

In both cases $|S| = 2$ for any deterministic strategy. Since any probabilistic classical strategy is a convex combination of deterministic strategies, and $|S|$ is a convex function of the strategy, it follows that $|S| \leq 2$ for all classical strategies.

F. Relationship Between S and the Winning Probability

The winning probability of a strategy is directly related to the CHSH parameter S. Since inputs are drawn uniformly at random, each pair (x,y) occurs with probability $1/4$, and the overall winning probability is:

$$P_{\text{win}} = \frac{1}{4} \sum_{x,y\,\in\,\{0,1\}} P(\text{win} \mid x,y). \tag{19}$$

In terms of signed variables, the winning condition $\tilde{a}\tilde{b} = (-1)^{x\cdot y}$ implies:

$$P(\text{win} \mid x,y) = \frac{1 + (-1)^{x\cdot y}\,E(x,y)}{2}. \tag{20}$$

Substituting (20) into (19) and summing explicitly over all input pairs:

$$P_{\text{win}} = \frac{1}{4}\left[\frac{1+E(0,0)}{2} + \frac{1+E(0,1)}{2} + \frac{1+E(1,0)}{2} + \frac{1-E(1,1)}{2}\right]$$
$$= \frac{1}{2} + \frac{E(0,0)+E(0,1)+E(1,0)-E(1,1)}{8}$$
$$= \frac{1}{2} + \frac{S}{8}, \tag{21}$$

where we have used the definition (14) in the final step. The classical bound $|S| \leq 2$ therefore implies directly:

$$P_{\text{win}}^{\text{classical}} \leq \frac{1}{2} + \frac{2}{8} = \frac{3}{4} = 75\%, \tag{22}$$

in complete agreement with the combinatorial argument presented in Section I A. The relation (21) further establishes that maximizing the winning probability is equivalent to maximizing S, providing the formal bridge between the game-theoretic and Bell inequality formulations of the CHSH problem.

G. Quantum Strategy: Projective Measurements and Quantum Correlations

1. Projective Measurements and the Born Rule

In the quantum formulation of the CHSH game, the binary outputs $a, b \in \{0,1\}$ of Alice and Bob are obtained as outcomes of projective measurements performed on a shared bipartite quantum state. A projective measurement on a qubit associated with a binary outcome $v \in \{0,1\}$ is described by a complete set of orthogonal projectors $\{\Pi_0, \Pi_1\}$ satisfying:

$$\Pi_0 + \Pi_1 = \mathbb{I}, \tag{23}$$
$$\Pi_v^2 = \Pi_v, \tag{24}$$
$$\Pi_v \Pi_{v'} = \delta_{vv'}\, \Pi_v, \tag{25}$$

where $\mathbb{I}$ denotes the single-qubit identity operator. By the Born rule, the probability of obtaining outcome v when measuring a state $|\psi\rangle$ is:

$$P(v) = \langle\psi|\Pi_v|\psi\rangle. \tag{26}$$

For a two-qubit state $|\Psi\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$, Alice performs a projective measurement $\{\Pi_0^A(\alpha), \Pi_1^A(\alpha)\}$ on her qubit at angle α, and Bob performs $\{\Pi_0^B(\beta), \Pi_1^B(\beta)\}$ on his qubit at angle β. The joint probability of outcomes (a, b) is:

$$P(a, b \mid \alpha, \beta) = \langle\Psi|\Pi_a^A(\alpha) \otimes \Pi_b^B(\beta)|\Psi\rangle. \tag{27}$$

2. Measurement Observables in the Sign Basis

To connect with the signed formulation of Section I B, we associate with each projective measurement the dichotomic observable:

$$\hat{A}(\alpha) = \Pi_0^A(\alpha) - \Pi_1^A(\alpha), \quad \hat{B}(\beta) = \Pi_0^B(\beta) - \Pi_1^B(\beta), \tag{28}$$

whose eigenvalues are $\{+1, -1\}$, corresponding precisely to the signed variables $\tilde{a}$ and $\tilde{b}$ introduced in (2). The quantum analog of the classical correlation function (9) is:

$$E_Q(\alpha, \beta) = \langle\Psi|\hat{A}(\alpha) \otimes \hat{B}(\beta)|\Psi\rangle$$
$$= \sum_{a,b \in \{0,1\}} (-1)^{a+b}\, P(a, b \mid \alpha, \beta). \tag{29}$$

This expression is the direct quantum translation of the classical correlation (9): the classical probability distribution $P(a, b \mid x, y)$ is replaced by Born rule probabilities, and the classical signed outputs by eigenvalues of dichotomic observables.

3. Explicit Form of the Projectors

For a qubit measured along the direction defined by angle θ in the XZ-plane of the Bloch sphere, the measurement eigenstates are:

$$|\psi_\theta\rangle = \cos\theta\,|0\rangle + \sin\theta\,|1\rangle, \tag{30}$$
$$|\psi_{\theta+\pi/2}\rangle = -\sin\theta\,|0\rangle + \cos\theta\,|1\rangle, \tag{31}$$

corresponding to eigenvalues $+1$ and -1 respectively. The associated projectors are:

$$\Pi_0(\theta) = |\psi_\theta\rangle\langle\psi_\theta| = \begin{pmatrix} \cos^2\theta & \cos\theta\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta \end{pmatrix}, \tag{32}$$

$$\Pi_1(\theta) = |\psi_{\theta+\pi/2}\rangle\langle\psi_{\theta+\pi/2}| = \begin{pmatrix} \sin^2\theta & -\cos\theta\sin\theta \\ -\cos\theta\sin\theta & \cos^2\theta \end{pmatrix}, \tag{33}$$

and the corresponding dichotomic observable is:

$$\hat{M}(\theta) = \Pi_0(\theta) - \Pi_1(\theta) = \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}. \tag{34}$$

One may verify that $\hat{M}(\theta)$ is Hermitian, has eigenvalues ± 1, and satisfies $\hat{M}(\theta)^2 = \mathbb{I}$, consistent with (23)–(25).

4. Quantum Correlation for the Bell State

We compute the quantum correlation function (29) for the maximally entangled Bell state:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}\big(|00\rangle + |11\rangle\big), \tag{35}$$

with Alice measuring at angle α and Bob at angle β. Expanding the expectation value explicitly:

$$\begin{aligned} E_Q(\alpha,\beta) &= \langle\Phi^+|\,\hat{M}(\alpha)\otimes\hat{M}(\beta)\,|\Phi^+\rangle \\ &= \frac{1}{2}\big(\langle 00| + \langle 11|\big)\big(\hat{M}(\alpha)\otimes\hat{M}(\beta)\big) \\ &\quad\times\big(|00\rangle + |11\rangle\big). \end{aligned} \tag{36}$$

Using the matrix elements of $\hat{M}(\theta)$:

$$\langle 0|\hat{M}(\theta)|0\rangle = \cos 2\theta, \tag{37}$$
$$\langle 1|\hat{M}(\theta)|1\rangle = -\cos 2\theta, \tag{38}$$
$$\langle 0|\hat{M}(\theta)|1\rangle = \sin 2\theta, \tag{39}$$
$$\langle 1|\hat{M}(\theta)|0\rangle = \sin 2\theta, \tag{40}$$

the four contributions to (36) evaluate as:

$$\langle 00|\hat{M}(\alpha)\otimes\hat{M}(\beta)|00\rangle = \cos 2\alpha\,\cos 2\beta, \tag{41}$$
$$\langle 11|\hat{M}(\alpha)\otimes\hat{M}(\beta)|11\rangle = \cos 2\alpha\,\cos 2\beta, \tag{42}$$
$$\langle 00|\hat{M}(\alpha)\otimes\hat{M}(\beta)|11\rangle = \sin 2\alpha\,\sin 2\beta, \tag{43}$$
$$\langle 11|\hat{M}(\alpha)\otimes\hat{M}(\beta)|00\rangle = \sin 2\alpha\,\sin 2\beta. \tag{44}$$

Summing all contributions with the factor 1/2:

$$\begin{aligned} E_Q(\alpha,\beta) &= \cos 2\alpha\cos 2\beta + \sin 2\alpha\sin 2\beta \\ &= \cos\big(2\alpha - 2\beta\big). \end{aligned} \tag{45}$$

This is a central result: the quantum correlation function for $|\Phi^+\rangle$ depends exclusively on the angular difference between Alice's and Bob's measurement settings.

H. Maximization of the Quantum CHSH Parameter

1. Quantum CHSH Parameter

In the quantum game, Alice measures at angle α_x for input $x \in \{0,1\}$, and Bob at angle β_y for input $y \in \{0,1\}$. Substituting (45) into (14):

$$\begin{aligned} S_Q = &\cos 2(\alpha_0 - \beta_0) + \cos 2(\alpha_0 - \beta_1) \\ &+ \cos 2(\alpha_1 - \beta_0) - \cos 2(\alpha_1 - \beta_1). \end{aligned} \tag{46}$$

2. Optimality Conditions

We seek angles $\{\alpha_0, \alpha_1, \beta_0, \beta_1\}$ maximizing S_Q. Setting $\alpha_0 = 0$ as reference and taking partial derivatives:

$$\frac{\partial S_Q}{\partial \beta_0} = 0 \implies \sin 2(\alpha_0 - \beta_0) + \sin 2(\alpha_1 - \beta_0) = 0, \tag{47}$$
$$\frac{\partial S_Q}{\partial \beta_1} = 0 \implies \sin 2(\alpha_0 - \beta_1) + \sin 2(\alpha_1 - \beta_1) = 0, \tag{48}$$
$$\frac{\partial S_Q}{\partial \alpha_1} = 0 \implies \sin 2(\alpha_1 - \beta_0) = \sin 2(\alpha_1 - \beta_1). \tag{49}$$

Applying the sum-to-product identity $\sin u + \sin v = 2\sin\frac{u+v}{2}\cos\frac{u-v}{2}$ to (47):

$$\sin\big(\alpha_0 + \alpha_1 - 2\beta_0\big)\cos\big(\alpha_0 - \alpha_1\big) = 0. \tag{50}$$

An identical analysis of (48) yields:

$$\sin\big(\alpha_0 + \alpha_1 - 2\beta_1\big)\cos\big(\alpha_0 - \alpha_1\big) = 0. \tag{51}$$

Assuming $\cos(\alpha_0 - \alpha_1) \neq 0$, equations (50) and (51) require:

$$\beta_0 = \frac{\alpha_0 + \alpha_1}{2}, \qquad \beta_1 = \frac{\alpha_0 + \alpha_1}{2} + \frac{\pi}{4}. \tag{52}$$

Substituting (52) into (49) with $\alpha_0 = 0$:

$$\sin\left(\frac{\alpha_1}{2}\right) = \sin\left(\frac{\alpha_1}{2} - \frac{\pi}{2}\right) \implies \alpha_1 = \frac{\pi}{2}, \tag{53}$$

from which, applying a global rotation of $-\pi/8$ to preserve angular differences, the optimal measurement angles are:

$$\boxed{\alpha_0 = 0, \quad \alpha_1 = \frac{\pi}{4}, \quad \beta_0 = \frac{\pi}{8}, \quad \beta_1 = -\frac{\pi}{8}.} \tag{54}$$

3. Tsirelson's Bound

Substituting (54) into (46):

$$\begin{aligned} S_Q^{\max} &= \cos\left(\frac{\pi}{4}\right) + \cos\left(\frac{\pi}{4}\right) \\ &\quad + \cos\left(\frac{\pi}{4}\right) - \cos\left(\frac{3\pi}{4}\right) \\ &= \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} \\ &= 2\sqrt{2} \approx 2.828, \end{aligned} \tag{55}$$

which is Tsirelson's bound[?] , strictly exceeding the classical bound $|S| \leq 2$. The corresponding optimal winning probability follows from (21):

$$P_{\text{win}}^{\text{quantum}} = \frac{1}{2} + \frac{2\sqrt{2}}{8} = \frac{2+\sqrt{2}}{4} \approx 85.36\%, \tag{56}$$

representing a quantum advantage of approximately 10.36 percentage points over the optimal classical strategy. Table II summarizes the theoretical predictions for both strategies.

TABLE II: Theoretical comparison between classical and quantum strategies for the CHSH game.

Strategy	S	P_{win}
Optimal classical	2	75.00%
Optimal quantum	$2\sqrt{2}$	85.36%